Unit – 2

Scenario based Asessment – Report

CSA1717 – Artificial Intelligence

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QUESTION 1 – AI-Powered Drone Delivery (Greedy & A\)*

1.1 Scenario Understanding

The government deploys AI-powered drones to deliver emergency medicines in a flood-affected region. The drone must navigate blocked roads, dynamic weather conditions, and varying terrain costs. It receives partial real-time updates and uses heuristic estimates to choose safe and fast routes.

1.2 Greedy Best-First Search Behavior

Greedy Best-First Search selects the node with the lowest heuristic value (straight-line distance). It focuses only on **h(n)** and ignores actual movement cost.

Strengths

- Very fast
- Low memory usage
- Works well when heuristic is accurate

Weaknesses

- Not optimal
- Easily misled by blocked paths
- Unsafe in dynamic environments

1.3 A* **Search Behavior

A* uses

$$f(n) = g(n) + h(n)$$

It balances actual movement cost (terrain, weather) and heuristic distance.

Strengths

- Optimal when heuristic is admissible
- Safer route selection
- Adapts better to dynamic updates

Weaknesses

- Slower than Greedy
- Higher memory usage

1.4 Impact of Heuristic Accuracy

- **Greedy:** Highly sensitive; wrong heuristic → wrong path
- A\:* Less sensitive; g(n) compensates for heuristic errors

1.5 Effect of Dynamic Changes

- **Greedy:** Fails when paths suddenly become blocked
- A\:* Recalculates f(n) and replans efficiently

1.6 Recommendation

A* is the most suitable algorithm because it balances **optimality, safety, and real-time adaptability**.

QUESTION 2 – Smart City Traffic Optimization (Hill Climbing & Metaheuristics)

2.1 Scenario Understanding

A smart city AI optimizes traffic signal timings to reduce waiting time, fuel consumption, and congestion. The search space is extremely large and dynamic.

2.2 Why Traditional Search Is Inefficient

- Huge state space (hundreds of intersections × timings)
- No fixed goal state
- Continuous optimization required
- Dynamic traffic patterns

2.3 Hill Climbing Behavior

Hill Climbing makes small local adjustments and moves to better neighboring states.

Limitations

- Gets stuck in local maxima
- Cannot escape plateaus
- Poor scalability

2.4 Real-World Issues

- **Local maxima:** One junction optimized but city-wide congestion remains
- **Plateaus:** Low traffic periods with no improvement
- **Ridges:** Requires long sequences of small changes

2.5 Overcoming Limitations

Simulated Annealing

- Accepts worse moves early
- Escapes local maxima
- Gradually stabilizes

Genetic Algorithms

- Population-based
- Crossover + mutation
- Highly scalable
- Adapts to dynamic traffic

2.6 Recommendation

Genetic Algorithms are best due to **scalability, adaptability, and robustness**.

QUESTION 3 – Mars Rover Exploration (Online Search)

3.1 Scenario Understanding

A Mars rover explores unknown terrain with limited sensing, hazards, and incomplete maps. It must make real-time decisions.

3.2 Why Online Search Is Required

Offline search requires a complete map, which is impossible on Mars. Online search interleaves **planning + execution + learning**.

3.3 Challenges

- Partial observability
- Unknown hazards
- Dynamic terrain
- Limited sensing range

3.4 Rover's Internal Model Updating

- Builds a belief map
- Marks safe zones, hazards, unknown areas
- Updates costs as new terrain is discovered
- Uses LRTA* or D* Lite for real-time replanning

3.5 Comparison with Other Searches

- **Uninformed:** Blind, inefficient
- **Informed:** Needs full map
- **Online:** Learns while exploring, adapts dynamically

3.6 Recommendation

LRTA / D\ Lite** are most suitable due to **uncertainty handling, adaptability, and safety**.

QUESTION 4 – Exam Timetable Scheduling (CSP)

4.1 Scenario Understanding

The university must generate exam timetables satisfying constraints like hall limits, faculty availability, no overlapping exams, and precedence rules.

4.2 CSP Formulation

Variables

- Exam slots for each course
- Hall assignments
- Faculty schedules

Domains

- Available time slots
- Available halls
- Valid faculty timings

Constraints

- No overlapping exams for students
- Hall capacity limits
- Precedence constraints
- Faculty availability
- Special accommodations

4.3 Backtracking Search

- Assign a variable

- Check constraints
- If conflict → backtrack
- Systematically explores valid assignments

4.4 Constraint Propagation (Forward Checking)

- Eliminates invalid future domain values
- Prevents exploring impossible paths
- Improves efficiency significantly

4.5 Real-World Challenges

- Large number of students and courses
- Many overlapping constraints
- Dynamic changes (faculty leave, hall unavailability)

4.6 Recommendation

Backtracking + Forward Checking + MRV heuristic Best for **scalability, efficiency, and constraint satisfaction**.

QUESTION 5 – Real-Time Strategy Game AI (Minimax & Alpha-Beta)

5.1 Scenario Understanding

A gaming company develops AI for a real-time strategy game with strict time limits and large decision trees.

5.2 Minimax Modeling

- Max player (AI) tries to maximize score
- Min player (human) tries to minimize AI score
- Assumes optimal opponent
- Explores game tree to choose best move

5.3 Computational Challenges

- Huge branching factor
- Deep game tree
- Real-time constraints
- Exponential complexity

5.4 Alpha-Beta Pruning

- Uses α and β bounds
- Prunes branches that cannot affect final decision
- Reduces computation drastically
- Allows deeper search within time limits

5.5 Evaluation Function Design

Should consider:

- Material advantage
- Map control
- Unit health
- Strategic positions
- Future threats

These factors directly influence winning probability.

5.6 Recommendation

Iterative Deepening + Alpha-Beta + Strong Evaluation Function Best balance of **optimality and real-time performance**.